

Web Development Internship Task Report

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November 18, 2025

Task Overview

Task Title: Make a Website Mobile-Friendly Using CSS Media Queries

Objective: Convert an existing desktop-only page layout into a mobile-friendly responsive design using CSS media queries.

Deliverable: An updated HTML/CSS file demonstrating responsiveness on mobile viewports (max-width: 768px).

1 Introduction

In modern web development, ensuring a website is accessible and visually coherent on mobile devices is critical. This task focuses on implementing "Responsive Design" principles. The goal was to take a standard desktop layout—featuring a horizontal navigation bar and side-by-side content columns—and adapt it for smaller screens so that elements stack vertically and remain readable.

2 Methodology

The solution involved the following steps:

1. **HTML Structure:** Created a semantic HTML5 structure containing a `<header>` with navigation, a `<section>` for the hero banner, and a main content area using `flexbox` to display three information cards.
2. **Desktop Styling:** Applied base CSS to create a horizontal layout suitable for wide screens (monitors/laptops).
3. **Mobile Adaptation (Media Queries):** Implemented the `@media (max-width: 768px)` rule to target mobile devices.

3 Implementation Details

The core logic for the mobile responsiveness relies on changing the `flex-direction` from `row` (default) to `column` when the screen width drops below 768 pixels.

CSS Code Snippet

Below is the key CSS code used to achieve the responsive behavior:

```

1  /* Base Row Style (Desktop) */
2  .row {
3      display: flex;
4      gap: 2rem;
5  }
6
7  /* Mobile Styles */
8  @media (max-width: 768px) {
9
10     /* Stack navigation links vertically */
11     nav ul {
12         flex-direction: column;
13         gap: 10px;
14         width: 100%;
15     }
16
17     /* Stack content columns vertically */
18     .row {
19         flex-direction: column;
20         gap: 1.5rem;
21     }
22
23     /* Adjust typography */
24     .hero h1 {
25         font-size: 1.8rem;
26     }
27 }

```

Listing 1: Media Query Implementation

4 Outcome

The implemented solution successfully meets the requirements:

- **Navigation:** On desktop, links appear in a horizontal line. On mobile, they stack vertically for easier tapping.
- **Content Layout:** The three feature cards sit side-by-side on large screens but stack one on top of the other on mobile devices to prevent horizontal scrolling.
- **Readability:** Font sizes and padding were adjusted to ensure the content is legible on smaller viewports.

5 Conclusion

This task demonstrated the importance of fluid layouts and media queries in web development. By utilizing simple CSS rules, we transformed a rigid desktop page into a flexible, mobile-friendly interface, ensuring a positive user experience across all devices.

Report submitted by Soumen Das